

Software IP Card

Basic Information		
Licensing		Open-source (SolderPad v0.52)
Target FPGA or ASIC		ASIC, VCU118, VCU128
Design Language		SystemVerilog
System-Level Features		
Coupling Type		Tightly-coupled
Data Types		INT8-64, FP16-64
ISA		RISC-V
Peak Ops Per Cycle		2 - 16 lanes (1 FPU each). On FPGA: 2 lanes = 2 DP ops / cycle
Interfaces		CV-X-IF, AXI
Supported Operations		RISC-V vector extension RVV 1.0
Compiler Support		OpenSource RISC-V toolchain (GCC or LLVM)
External Memory Required		optional; L2 / DDR
Cache Or Scratchpad		AXI compliant: Cache + Scratchpad
Architecture		
Type		Von Neumann, SMP
Num Clock Domains		1
Control Interface	ISAExtension	RISC-V V v1.0
Control Interface	Memory Mapped	No
Control Interface	Protocol	CV-X-IF + MMU + coherence IF
Memory Interface	Interface Protocol	AXI4
Memory Interface	Hierarchy Level	L2 (CVA6 L1 bypassed)
Memory Interface	Cached	
Memory Interface	IOMMU	
Interrupts		Yes
Microarchitecture		
Parametrization	Parametric Num Units	No
Parametrization	Parametric Config	Yes (# Vector Lanes, Vector Register Length)
Programmability	Programmable Cores	Yes (CVA6)
Programmability	ISAExtension	RISC-V (RV64 GCV)
Software		
Compiler	Requires Specialized Compiler	No (RISC-V compatible)
Compiler	Supported Compilers	
Hardware Abstraction Layer	HAL Functions	
highLevelAPI	Has High Level API	Yes (C intrinsics)
highLevelAPI	SDK Repository	

highLevelAPI	Has Domain Specific Compiler	
Integration		
IP Distribution	Manifest Type	Bender.yml
IP Distribution	Standalone Simulation	Yes
IP Distribution	Standalone Simulation SW Require-ments	QuestaSim or Verilator
IP Distribution	Integration Docs Or Examples	Yes
Synthesis	Is Synthesizable	Yes
Synthesis	FPGA Synthesis Examples	Yes
Synthesis	ASIC Synthesis Examples	Yes (proprietary)
Simulation	Closed Source Simulation	Yes (QuestaSim)
Simulation	Open Source Simulation	Yes (Verilator)
Evaluation	PPA Results Available	Yes
Physical Implementation		
ASIC Area	Num Gates	1.7 MGE (2 lanes) - 12.8 MGE (16 lanes)
ASIC Area	On Chip Internal Memory Size	
ASIC Area	Target Clock Frequency	>1 GHz (GF22nm), crit. path ca. 40 FO4 gates